Stochastic Processes Semester 1

Tutorial 9 Extension Questions – Solutions

1. (a) The jump chain is:

$$P^* = \begin{pmatrix} 0 & 1/4 & 1/4 & 1/4 & 1/4 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1/3 & 0 & 0 & 0 & 1/3 & 1/3 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

So there is one open class $\{1,3\}$ and 2 closed classes $C_1 = \{2,4\}$ and $C_2 = \{5,6\}$. Hence we have:

$$\alpha_1^{(C_1)} = 1/4\alpha_3^{(C_1)} + 1/2$$

and

$$\alpha_3^{(C_1)} = 1/3\alpha_1^{(C_1)}.$$

So $\alpha_1^{(C_1)} = 6/11$ and $\alpha_3^{(C_1)} = 2/11$, and thus $\alpha_1^{(C_2)} = 5/11$ and $\alpha_3^{(C_2)} = 9/11$.

(b) They are not same (Hint: Similar to Tutorial 9 Question 2 (b), this time $q_{22} \neq q_{44}$ and $q_{55} \neq q_{66}$.)

(c) The limiting distributions are:

i. $(0, 2/3, 0, 1/3, 0, 0)$

ii. $(0, 4/33, 0, 2/33, 18/55, 27/55)$

2. (a) Since $p_0 = 0$, the expected time to reach 0 is ∞.

(b) $\mathbf{E}\tau = 1/p = 2$

3. (a) $\mathbf{E}\tau = 1/p_2 = 5/3$

(b) The expected time to exceed the level 3 is 25/21. (Hint: The process stays at level 2 if each individual produces exactly one offspring.)

(c)

$$\mathbf{E}T = \mathbf{E}\tau_1 + \frac{4}{7}\mathbf{E}\tau_2 = \frac{25}{21} + \frac{4}{7} \cdot \frac{125}{117} = \frac{1475}{819}.$$

(Hint: According to part (a), the first time τ_1 when $X_{\tau_1} > 2$ has a geometric distribution with parameter $p = 21/25$. At time τ_1, the probability of jumping to level 4 is 3/7, where the process stops. Otherwise (with probability 4/7) there is a jump to level 3 and then the process reaches set $\{4, 5, 6, \ldots\}$ after a further geometric random variable τ_2 with parameter $p = 1 - (2/5)^3 = 117/125$.)